This assignment will not be graded and is only for practice.

Key Points: 1

A matrix is in row echelon form if :

• The first non-zero element (the leading entry) in a row is 1.

• The column containing the leading 1 of a row is to the right of the column containing the leading 1 of the row above it.

• Any non-zero rows are always above rows with all zeros.

A matrix is in reduced row echelon form if :

• The first non-zero element in the first row (the leading entry) is the number 1.

• All subsequent non-zero rows must also have their leading entries (i.e. first non-zero entries) as 1 and they should appear to the right of the leading entry in the previous row.

• The leading entry in each row must be the only non-zero number in its column.

• Any non-zero rows are always above rows with all zeros.

Note: Any matrix which is in reduced row echelon form is also in row echelon form.

1) Choose the correct options for the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

1 point

[Hint: Find out the leading entries for each rows and observe whether the matrix is satisfying the conditions of row echelon form and reduced row echelon form.]

- ◯ A is in row echelon form but not in reduced row echelon form.
- ◯ A is in both row echelon form and reduced row echelon form.
- ◯ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is in both row echelon form and reduced row echelon form.

2) Choose the correct options for the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

1 point

- ◯ A is in row echelon form but not in reduced row echelon form.
- ◯ A is in both row echelon form and reduced row echelon form.
- ◯ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is neither in row echelon form nor reduced row echelon form.

3)

1 point

Choose the correct options for the matrix $F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- ◯ A is in row echelon form but not in reduced row echelon form.
- ◯ A is in both row echelon form and reduced row echelon form.
- ◯ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is in row echelon form but not in reduced row echelon form.

Key Points: 2

Let $Ax = b$ be a system of linear equations and suppose A is in reduced row echelon form. Assume that for every zero row of A, the corresponding entry of b is also 0 (i.e., if the $i - th$ row of A is 0, then so is b_i.)

- If the $i - th$ column has the leading entry of some row, we call x_i a dependent variable.
- If the $i - th$ column does not have the leading entry of any row, we call x_i an independent variable.

4) Consider a system of linear equations: **1 point**

$$0x_1 + x_2 + 0x_3 + 0x_4 = 1$$
$$0x_1 + 0x_2 + x_3 + 0x_4 = 1$$

Choose the set of correct options.
[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

- [ ] x_1 and x_2 are dependent variables.
- [ ] x_2 is a dependent variable.
- [ ] x_3 and x_4 are independent variables.
- [ ] x_4 is an independent variable.

No, the answer is incorrect.
Score: 0

Accepted Answers:

x_2 is a dependent variable.
x_4 is an independent variable.

5) **1 point**

let $Ax = b$ be a system of linear equations, where $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$.

Choose the set of correct options.

- [ ] x_1 is a dependent variable.
- [ ] x_2 is a dependent variable.
- [ ] x_3 is a dependent variable.
- [ ] The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

No, the answer is incorrect.
Score: 0

Accepted Answers:

x_1 is a dependent variable.
x_3 is a dependent variable.
The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

Key Points: 3

The three different types of elementary row operations that can be performed on a matrix are:

- **Type 1:** Interchanging two rows.
- **Type 2:** Multiplying a row with some constant.
- **Type 3:** Adding a scalar multiple of a row to another row.

Using these row operations one can obtain row echelon form or reduced row echelon form of a given matrix.

Consider the following matrix A and answer the questions (a), (b), (c), (d), and (e).

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 2 & 4 & 4 & 3 \end{bmatrix}$$

6) What will be the third row of the matrix if we perform the row operation $R_3 - 2R_1$ on A? ***1 point***

- $\begin{pmatrix} 1 & 1 & 2 & 3 \end{pmatrix}$
- $\begin{pmatrix} 4 & 10 & 8 & 3 \end{pmatrix}$
- $\begin{pmatrix} 0 & -2 & 0 & 3 \end{pmatrix}$
- $\begin{pmatrix} 0 & 4 & 4 & 3 \end{pmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{pmatrix} 0 & -2 & 0 & 3 \end{pmatrix}$

7) What will be the second row of the matrix if we perform the row operation $\frac{1}{2}R_2$ on A ? ***1 point***

- $\begin{pmatrix} 0 & 1 & 1 & 0 \end{pmatrix}$
- $\begin{pmatrix} 0 & 4 & 2 & 0 \end{pmatrix}$
- $\begin{pmatrix} 0 & 1 & \frac{1}{2} & 0 \end{pmatrix}$
- $\begin{pmatrix} 1 & 2 & 2 & \frac{3}{2} \end{pmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{pmatrix} 0 & 1 & \frac{1}{2} & 0 \end{pmatrix}$

8) If we perform the row operations mentioned in question (a) and question (b), simultaneously on A, the matrix (let it be called B) we get is: ***1 point***

- $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 4 & 4 & 3 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$

9) Which row operations have to be performed on the matrix B (matrix B is as in question (c)) to make all the entries 0 in the second column, except the leading entry of the second row. ***1 point***

- $R_1 - 3R_2$ and $R_3 - 2R_2$
- $R_1 + 3R_2$ and $R_3 + 2R_2$
- $R_1 + 3R_2$ and $R_3 - 2R_2$
- $R_1 - 3R_2$ and $R_3 + 2R_2$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$R_1 - 3R_2$ and $R_3 + 2R_2$

10) The reduced row echelon form of A is **1 point**

[Hint: Recall the definition of reduced row echelon form.]

- ○ $\begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$
- ○ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$
- ○ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$
- ○ $\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$

Key Points: 4

Let $Ax = b$ denote the matrix representation of a system of linear equations.
The augmented matrix is denoted by $[A|b]$.
The matrix obtained by performing the operations that transform A to its reduced row echelon form R on the augmented matrix is denoted by $[R|c]$.
The solutions of $Rx = c$ are the same as the solutions of $Ax = b$.

Consider a system of linear equations

$$2x_1 + x_2 = 1$$
$$-x_1 + x_3 + x_4 = -1$$
$$x_1 + x_2 - x_3 + x_4 = 2$$
$$-x_1 + x_3 + x_4 = 1.$$

Answer the questions (a), (b) and (c) based on the above information.

11) Which of the following represents an augmented matrix of the system? ***1 point***

[Hint: First try to write down the coefficient matrix A and the corresponding vector b.]

- $\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 1 & 0 \\ 1 & 1 & -1 & 1 & 0 \\ -1 & 0 & 1 & 1 & 0 \end{array}\right]$

- $\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 0 & 1 & 1 & 1 \end{array}\right]$

- $\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & -1 \\ -1 & 0 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 0 & 1 & 1 & -1 \end{array}\right]$

- $\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & -1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 1 & 1 & 0 & -1 \end{array}\right]$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$$\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 0 & 1 & 1 & 1 \end{array}\right]$$

12) What will be the matrix $[R\,|\,c]$ obtained by performing the operations that transform A to its reduced row echelon form R on the augmented matrix? **1 point**

- ○ $$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right]$$

- ○ $$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right]$$

- ○ $$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

- ○ $$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right]$$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right]$$

13) Choose the correct option. [Hint: Write down the system of linear equations corresponding to the reduced row echelon form of the augmented matrix.] ***1 point***

- The system of linear equations has an infinite number of solutions.
- The system of linear equations has a unique solution.
- The system of linear equations has no solution.
- $x_1 = x_2 = x_3 = x_4 = 1$ is a solution of the system of linear equations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has no solution.

Your score is: 0/13